

DOPPELGANGER.WORLD

END-TO-END ARCHITECTURE & TECH STACK SPECIFICATION

System Architecture & Complete Technology Specification Guide

1. Executive Overview & Application Architecture

Doppelganger.world is a high-performance, web-based AI facial recognition and double-discovery platform. The application empowers users to discover their real-life look-alikes across a global community gallery, set bounty rewards for spotters, and execute 100% free client-side AI facial similarity scans directly inside the browser canvas.

The system is architected as a hybrid Serverless/Node.js full-stack system emphasizing high performance, zero-latency client-side AI vector extraction, low memory footprint, and robust multi-tier persistent cloud storage.

2. End-to-End Workflow & Pipeline

- **Phase 1: User Onboarding & Auth:** Secured via Clerk Auth supporting OAuth, email verification, and session state.
- **Phase 2: Photo Upload & Client-Side AI Pre-Validation:** Users select a face image. Before sending network bytes, client-side TinyFaceDetector verifies that a valid human face exists.
- **Phase 3: 128-D Vector Embedding Extraction:** The browser executes faceRecognitionNet (face-api.js) to compute a 128-dimensional floating-point feature embedding.
- **Phase 4: Node.js Express Backend Ingestion:** Multipart form data is parsed via Multer memory storage. Cloudinary API uploads and optimizes media delivery; Supabase PostgreSQL records profile metadata.
- **Phase 5: Community Gallery & Bounty Network:** Submissions appear in an interactive glassmorphic grid with multi-currency bounty rewards (INR, USD, EUR, GBP).
- **Phase 6: AI Doppelganger Matching Engine:** Framer-style Lightbox Modal executes client-side similarity comparison against stored database embeddings using Euclidean distance metrics.

3. Complete Technology Stack (Start to End)

3.1 Frontend & User Interface Layer

- **Core Stack:** Vanilla HTML5, Modern CSS3 with CSS Custom Properties (Variables), Framer Spotlight Animations, Glassmorphism.
- **Client-Side Logic:** ES6+ Modular JavaScript (script.js) with zero heavy framework overhead for maximum rendering speed.
- **Authentication API:** Clerk Auth (@clerk/clerk-js) handling user sign-in/up, user profiles, and secure admin interfaces.
- **Analytics & Monitoring:** Google Analytics 4 (GA4), Microsoft Clarity for heatmaps and session replay recording.
- **Monetization API:** Google AdSense integration (Publisher ID: ca-pub-7208166578795598).

3.2 Client-Side AI Computer Vision Engine

- **Library Engine:** @vladmandic/face-api (v1.0.2) loaded dynamically via JSDelivr CDN.
- **Neural Net Models:** TinyFaceDetector (Face detection), FaceLandmark68TinyNet (68 Landmark extraction), FaceRecognitionNet (128-D embedding computation).
- **Privacy & Efficiency:** AI inference runs 100% inside the user browser WebGL/Canvas, minimizing server GPU costs and protecting privacy.

3.3 Backend Infrastructure & Application APIs

- **Runtime Environment:** Node.js (v24 ES Modules) executing server.js Express application.
- **Web Server Framework:** Express.js (v4.21.2) with compression (Gzip Level 6) and CORS middleware.
- **File Upload Handler:** Multer (v1.4.5-lts.1) with memory storage buffer (10MB file limit).
- **Generative AI Chat API:** Groq Cloud API (api.groq.com/openai/v1/chat/completions) using Google Auth Library for assistant interactions.
- **SEO Compiler System:** Custom SSG engine (compile_pages.js, compile_all.js, generate_blogs.js) generating 75+ SEO static pages with Schema.org JSON-LD.

3.4 Cloud Database & Storage Services

- **Media Storage API:** Cloudinary API (v1.41.3) for automated image compression, webp conversion, and global CDN delivery.
- **Database API:** Supabase Cloud PostgreSQL (@supabase/supabase-js v2.55.0) storing user profiles, bounty rewards, currencies, and match records.
- **Local Fallback Cache:** Resilient local data.json store ensuring 100% uptime even during network or database connectivity events.

4. Complete API Specification & Data Contracts

Backend Rest APIs (Express & Node.js)

- GET /api/images — Fetches deduplicated image gallery combining Supabase DB & local data.json
- POST /api/upload — Ingests photo, name, email, currency & bounty (min 100). Uploads to Cloudinary & Supabase
- GET /api/blogs — Retrieves paginated SEO blog post metadata catalog
- GET /api/blogs/:slug — Fetches detailed Markdown/HTML blog content by slug
- DELETE /api/images/:id — Admin-authenticated endpoint for gallery content moderation

External Cloud & Third-Party APIs

- Cloudinary REST API — Uploads image binary buffers, returns secure CDN HTTPS URLs
- Supabase REST/PostgREST API — Direct PostgreSQL CRUD queries via service key
- Groq OpenAI-Compatible Chat API — Real-time AI chat stream for platform assistance
- Clerk Authentication API — Handles OAuth 2.0 JWT token generation and user session validation
- JSDelivr CDN API — Delivers quantized neural network weight models for face-api.js

5. Mathematics of AI Facial Matching Engine

Facial similarity matching in Doppelganger.world relies on 68 facial landmark detection and 128-dimensional vector embedding comparison. The 128-D descriptor vectors (u and v) represent normalized facial geometric features.

- **Euclidean Distance Metric:** Calculates geometric separation in 128-D space: $d(u, v) = \sqrt{\sum (u_i - v_i)^2}$.
- **Similarity Percentage Mapping:** Maps Euclidean distance (where $d < 0.65$ indicates strong match) to human-readable percentage: $S(d) = \max(70, \min(99.6, (1 - d/0.65) * 100))\%$.
- **Landmark Keypoints:** Measures interpupillary distance, eye depth, nose ridge, jawline curvature, and lip boundaries.

6. Summary Table of Tech Stack Components

Layer	Technology / API	Package / Version	Primary Purpose
Frontend UI	HTML5 / CSS3 / ES6 JS	Native Browser	Responsive UI & Animations
Auth API	Clerk Auth	@clerk/clerk-js	User Sign-in / OAuth / Session
Client AI	face-api.js	v1.0.2 (JSDelivr)	68 Landmarks & 128-D Vector Embeddings
Backend App	Express.js / Node.js	v4.21.2 / v24 Node	REST API Routes & Form Processing
Image CDN	Cloudinary API	v1.41.3	Cloud Storage & Image Optimization
Database	Supabase PostgreSQL	@supabase/supabase-js v2.55	Persistent Storage of Profiles & Reputations
AI Chat API	Groq API	OpenAI Schema	Generative Assistant Chat
Analytics	GA4 & MS Clarity	Cloud Script Tag	User Session Replay & Telemetry